

Lecture 2 : Calculus and Linear Algebra

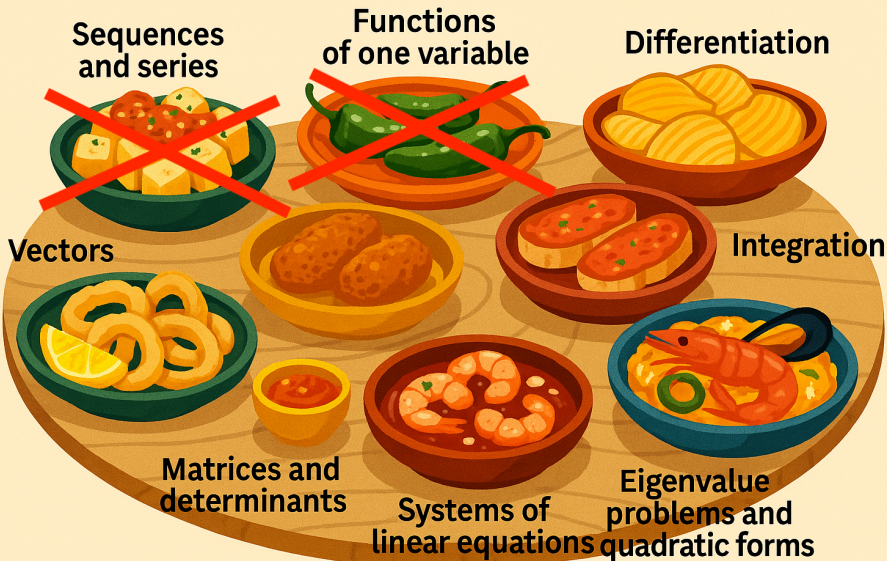
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Course content (math tapas)



Optimization in 1D

Monotonicity via derivatives

If f is differentiable on (a, b) :

1. f increasing on $[a, b] \iff f'(x) \geq 0$ on (a, b) .
2. f decreasing on $[a, b] \iff f'(x) \leq 0$ on (a, b) .
3. f constant on $[a, b] \iff f'(x) = 0$ on (a, b) .
4. $f'(x) > 0$ on $(a, b) \Rightarrow f$ strictly increasing on $[a, b]$.
5. $f'(x) < 0$ on $(a, b) \Rightarrow f$ strictly decreasing on $[a, b]$.

Converse of 4-5 can fail at isolated points (e.g., x^3).

Example: $f(x) = \frac{x^3}{3} + 2x^2 + 3x + 1$; $f'(x) = (x+1)(x+3)$ determines monotonicity intervals and local extrema.

Optimal points: first-order conditions

Local optima

Let $f : D \rightarrow \mathbb{R}$. A point $x_0 \in D$ is a **local maximum** if $\exists (a, b) \subset D$ with $x_0 \in (a, b)$ and $f(x) \leq f(x_0)$ for all $x \in (a, b)$. Similarly for **local min** with $\geq$.

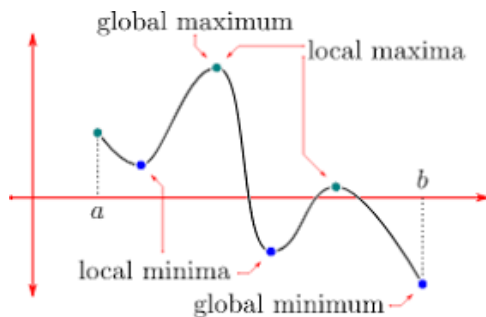
If f has a local optimum at interior x and is differentiable, then $f'(x) = 0$.

To see why this is true, stare at $f(x_0 + h) - f(x_0) \approx f'(x_0)h$.

This condition is **necessary**, not sufficient; see e.g. x^3 .

If x is stationary ($f'(x) = 0$) and f is strictly increasing to the left and strictly decreasing to the right, then x is a local max (reverse for min).

Optimal points



Example: Timing a sale (present value)

Suppose an assets market value is $V(t)$ at time t . At a constant annual interest rate R , the **present value** of selling at time t is

$$P(t) = V(t) e^{-Rt},$$

where e^{-Rt} discounts future cash flows.

Goal: Choose t to maximize $P(t)$.

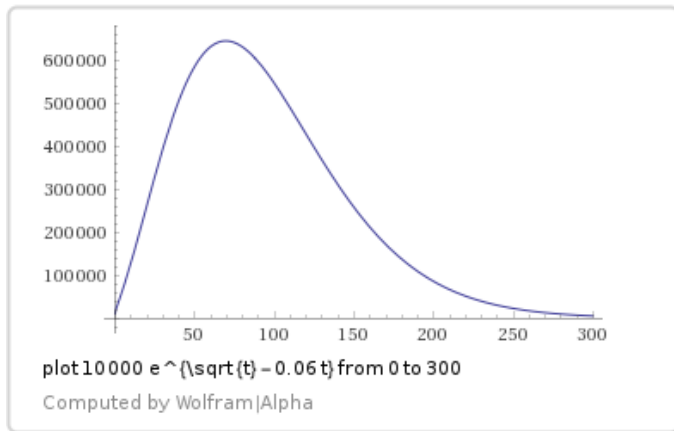
FOC (set derivative to zero):

$$P'(t) = V'(t)e^{-Rt} - R V(t)e^{-Rt} = 0 \quad \Rightarrow \quad \frac{V'(t)}{V(t)} = R.$$

Interpretation: Sell when the assets *instantaneous percentage growth rate* (matches) the *interest rate you could earn in the bank*.

Example: If $V(t) = 10000 e^{\sqrt{t}}$ and $R = 6\%$, then $P(t) = 10000 e^{\sqrt{t} - 0.06 t}$, which is maximized near $t \approx 69.44$.

Plot of the present value $P(t)$



To certify a **global** optimum on an interval, compare values at boundaries and critical points. Here $P(0) = 10000$, $\lim_{t \rightarrow \infty} P(t) = 0$.

Second-order conditions

Test: If $f'(x_0) = 0$ and $f''(x_0) < 0$ (> 0), then x_0 is a strict local max (min).

Indeed,
$$f(x_0 + h) \approx f(x_0) + f'(x_0)h + \frac{1}{2}f''(x_0)h^2 = f(x_0) + \frac{1}{2}f''(x_0)h^2.$$

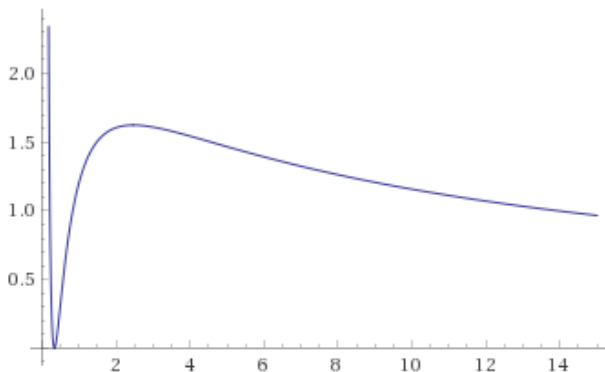
Example: For $f(x) = \frac{\log^2(3x)}{x}$ ($x > 0$):

$$f'(x) = \frac{(2 - \log 3x) \log 3x}{x^2}, \quad f''(x) = \frac{2(1 - 3 \log 3x + \log^2 3x)}{x^3}.$$

Critical points at $x_1 = \frac{1}{3}$ (min), $x_2 = \frac{e^2}{3}$ (max).

Question: What if $f''(x_0) = 0$? (assume all derivatives exist)

Plot of $f(x) = \frac{\log^2 3x}{x}$, $x > 0$



plot $\ln^2(3x)/x$ from 0 to 15 | Computed by Wolfram|Alpha

$\lim_{x \rightarrow \infty} f(x) = 0$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $f(x_1) = 0$, $f(x_2) > 0 \Rightarrow x_1$ global min,
no global max.

Convexity and concavity

If f is twice differentiable on (a, b) :

1. f convex on $[a, b] \Leftrightarrow f''(x) \geq 0$ for all $x \in (a, b)$.
2. f concave on $[a, b] \Leftrightarrow f''(x) \leq 0$ for all $x \in (a, b)$.
3. $f'' > 0$ (< 0) $\Rightarrow f$ strictly convex (concave).

Theorem: If f is convex and nondecreasing, and g is convex, then $f \circ g$ is convex. If f is convex and nonincreasing, and g is concave, then $f \circ g$ is convex.

Example: $f(x) = e^{x^2}$ is strictly convex on $[0, \infty)$.

Economics:

- **Cost functions:** usually convex — producing extra units becomes progressively more expensive (e.g. overtime wages, machine wear).
- **Utility functions:** often concave — reflects diminishing marginal utility (the first slice of pizza gives more satisfaction than the tenth).
- **Machine learning:** convex losses (e.g. squared error, logistic loss + ℓ_2 regularization) ensure optimization problems are tractable and stable.

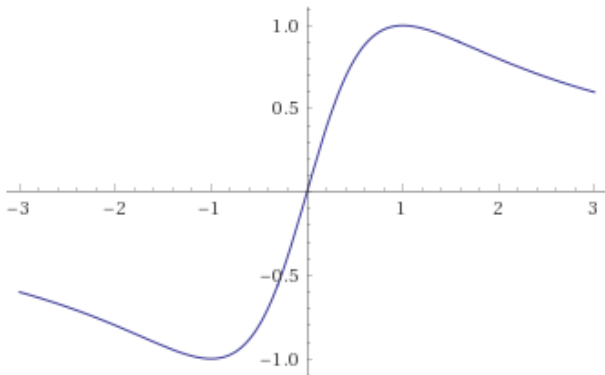
Example of changing curvature

Let $f(x) = \frac{2x}{x^2 + 1}$. Then

$$f'(x) = \frac{2(1 - x^2)}{(x^2 + 1)^2}, \quad f''(x) = \frac{4(x^3 - 3x)}{(x^2 + 1)^3}.$$

So f is strictly convex on $[-\sqrt{3}, 0] \cup [\sqrt{3}, \infty)$ and strictly concave on $(-\infty, -\sqrt{3}] \cup [0, \sqrt{3}]$.

Plot of $f(x) = \frac{2x}{x^2 + 1}$



plot $2x/(x^2+1)$ from -3 to 3 | Computed by Wolfram|Alpha

Inflexion points at $x \in \{-\sqrt{3}, 0, \sqrt{3}\}$. Global max at $x = 1$, min at $x = -1$, and $\lim_{|x| \rightarrow \infty} f(x) = 0$.

Inflexion points

If f is C^n on (a, b) , an **inflexion point** at x_0 occurs when curvature changes sign. A sufficient test:

$$f''(x_0) = \dots = f^{(n-1)}(x_0) = 0, \quad f^{(n)}(x_0) \neq 0, \quad n \text{ odd.}$$

Example: $f(x) = x^3$ has $f''(0) = 0$, $f^{(3)}(0) = 6 \neq 0 \Rightarrow$ inflexion at 0.

Taylor series expansion

Mean-value theorem

Mean Value Theorem

If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad (\text{equivalently } f(b) = f(a) + f'(c)(b - a)).$$

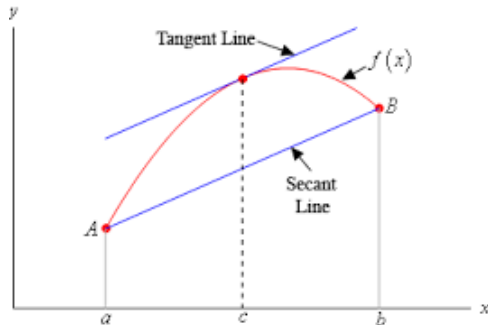
Proof idea: consider $h(x) = (f(b) - f(a))x - (b - a)f(x)$; since $h(a) = h(b)$, h has an interior extremum.

Interpretation: the instantaneous slope matches the secant slope at some point.

Geometry of the mean-value theorem

Recall: $\exists c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Taylor formula

Theorem (Taylor's formula of order n): Let f be $n + 1$ times differentiable on (a, b) , and let $x_0 \in (a, b)$ given. Then, for any $x \in (a, b)$, if $n = 1$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \text{error}.$$

If $n \geq 2$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + \text{error}.$$

The error is small if x is close to x_0 , therefore, this theorem says that differentiable functions may be **locally approximated by a polynomial** called the **Taylor polynomial**.

Taylor remainder and a handy expansion

Remainder (Lagrange form)

$$R_n^y(x) = \frac{f^{(n+1)}(y)}{(n+1)!} (x - x_0)^{n+1} \quad \text{for some } y \text{ between } x_0 \text{ and } x.$$

If $|f^{(n+1)}| \leq M$ near x_0 , then $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$.

Example (log around 1): With $x_0 = 1$,

$$\log x \approx (x - 1) - \frac{1}{2}(x - 1)^2, \quad R_2^y(x) = \frac{1}{3y^3}(x - 1)^3.$$

Short comment about small-o notation

Writing $|R_n^y(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}$ we conclude

$$\frac{R_n^y(x)}{|x - x_0|^n} \leq \frac{M}{(n+1)!} |x - x_0| \text{ and so}$$

$$\lim_{x \rightarrow x_0} \frac{R_n^y(x)}{|x - x_0|^n} = 0.$$

This we write as $R_n^y(x) = o(|x - x_0|^n)$.

In particular: If f is twice differentiable:

$$f(x+h) = f(x) + f'(x)h + o(|h|).$$

The generalization of this last expression to multivariate functions will provide powerful insights into their optimization.

Chapter 4: Integration

Often a function f is given and we are looking for F whose derivative is f .

Example: the marginal cost function $C'(x)$ is known (how cost changes with production x). We want the cost $C(x)$ itself.

Idea: **integration** reverses differentiation: it accumulates small changes.

Read Chapter 5 of Werner-Sotskov **Exercises** 5.1(a)-(b), 5.2(a)-(b), 5.3(c)

Indefinite integrals

Indefinite integrals

A differentiable F is an **antiderivative** of f if $F'(x) = f(x)$ on a common domain.

Fact: all antiderivatives differ by a constant: if $F'(x) = f(x)$, then any $\tilde{F}(x) = F(x) + c$ also satisfies $\tilde{F}'(x) = f(x)$.

Definition: the **indefinite integral** is

$$\int f(x) dx = F(x) + c.$$

Linearity:

1. $\int (f + g) dx = \int f dx + \int g dx$
2. $\int c f dx = c \int f dx$

Indefinite integrals you should know

Templates:

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$$

$$2. \int \frac{1}{x} dx = \log |x| + c$$

$$3. \int e^x dx = e^x + c$$

$$4. \int \sin x dx = -\cos x + c$$

$$5. \int \cos x dx = \sin x + c$$

$$6. \int a^x dx = \frac{a^x}{\log a} + c \quad (a > 0, a \neq 1)$$

Always check by differentiating the right-hand side.

Integration by substitution

Theorem (Substitution)

If $t = g(x)$ and F is an antiderivative of f , then

$$\int f(g(x)) g'(x) dx = \int f(t) dt = F(t) + c = F(g(x)) + c.$$

(if $F'(t) = f(t)$ then $f(g(x))g'(x) = (F(g(x)))'$ by the chain rule)

Examples:

$$1. \int (ax + b)^n dx = \frac{1}{a} \int t^n dt = \frac{(ax + b)^{n+1}}{a(n+1)} + c.$$

($t = ax + b$, so $dt = a dx$.)

$$2. \int \frac{e^x}{\sqrt[3]{1 + e^x}} dx = \int t^{-1/3} dt = \frac{3}{2} t^{2/3} + c = \frac{3}{2} (1 + e^x)^{2/3} + c.$$

($t = 1 + e^x$, so $dt = e^x dx$.)

Integration by parts

Theorem (Integration by parts)

For differentiable u, v ,

$$\int u(x)v'(x) dx = u(x)v(x) - \int u'(x)v(x) dx.$$

Proof: Differentiate $u(x)v(x)$ and integrate.

Example: with $u = \log x$, $v' = 1$,

$$\int \log x dx = x \log x - \int 1 dx = x(\log x - 1) + c.$$

A combo: substitution then parts

Compute $\int \sin \sqrt{x} \, dx$.

Substitute $t = \sqrt{x}$, so $x = t^2$ and $dx = 2t \, dt$:

$$\int \sin \sqrt{x} \, dx = 2 \int t \sin t \, dt.$$

Now parts with $u = t$, $v' = \sin t$:

$$2 \int t \sin t \, dt = 2 \left(-t \cos t + \int \cos t \, dt \right) = 2(-t \cos t + \sin t) + c.$$

Back-substitute $t = \sqrt{x}$:

$$\boxed{\int \sin \sqrt{x} \, dx = 2 \left(-\sqrt{x} \cos \sqrt{x} + \sin \sqrt{x} \right) + c}.$$

A combo: substitution then parts



Definite integrals

Definite integral as area and accumulation

For continuous $f : [a, b] \rightarrow \mathbb{R}$ with $f \geq 0$, the definite integral

$$\int_a^b f(x) dx$$

is the area under f between a and b . It accumulates signed change.

Properties:

1. $\int_b^a f = -\int_a^b f$
2. $\int_a^b cf = c \int_a^b f$
3. If $c \in [a, b]$, then $\int_a^b f = \int_a^c f + \int_c^b f$
4. $\left| \int_a^b f \right| \leq \int_a^b |f|$
5. If $f \leq g$ on $[a, b]$, then $\int_a^b f \leq \int_a^b g$

Some uses: cumulative revenue from a known marginal revenue curve, energy used by a device with power draw $P(t)$, or probability mass from a density.

Fundamental Theorem of Calculus

Fundamental Theorem of Calculus

If f is continuous on $[a, b]$ and F is an antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Moreover, $G(t) = \int_a^t f(x) dx$ is differentiable and $G'(t) = f(t)$.

Marginal to total: If $C'(x)$ is marginal cost, then the change in total cost from $x = 300$ to $x = 400$ is

$$C(400) - C(300) = \int_{300}^{400} C'(x) dx.$$

Example: $C'(x) = 6 - \frac{60}{x+1}$ for $x \in [0, 1000]$:

$$\int_{300}^{400} \left(6 - \frac{60}{x+1}\right) dx = \left(6x - 60 \log |x+1|\right)_{300}^{400} \approx 582.79.$$

Application: proving $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ via integrals

Define $\log x = \int_1^x \frac{1}{t} dt$. Let e^x be the inverse of $\log x$; then $1 = \int_1^e \frac{1}{t} dt$.

For $t \in [1, 1 + \frac{1}{n}]$,

$$\frac{1}{n+1} = \int_1^{1+\frac{1}{n}} \frac{1}{1+\frac{1}{n}} dt \leq \int_1^{1+\frac{1}{n}} \frac{1}{t} dt \leq \int_1^{1+\frac{1}{n}} 1 dt = \frac{1}{n}.$$

So

$$\frac{1}{n+1} \leq \log \left(1 + \frac{1}{n}\right) \leq \frac{1}{n}.$$

Exponentiate and rearrange to obtain

$$\frac{e}{1 + \frac{1}{n}} \leq \left(1 + \frac{1}{n}\right)^n \leq e,$$

and let $n \rightarrow \infty$.



Application: Taylor with integral remainder (up to $n = 2$)

Apply Fundamental Theorem of Calculus repeatedly:

$$\begin{aligned}f(x) &= f(x_0) + \int_{x_0}^x f'(t_1) dt_1 \\&= f(x_0) + f'(x_0)(x - x_0) + \int_{x_0}^x \int_{x_0}^{t_1} f''(t_2) dt_2 dt_1 \\&= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \int_{x_0}^x \int_{x_0}^{t_1} \int_{x_0}^{t_2} f^{(3)}(t_3) dt_3 dt_2 dt_1.\end{aligned}$$

By the intermediate value theorem there is y between x_0 and x with

$$\int \int \int f^{(3)}(t_3) dt_3 dt_2 dt_1 = \frac{f^{(3)}(y)}{3!}(x - x_0)^3.$$

Chapter 5: Vectors

A **vector** is an ordered n -tuple of real numbers: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

Why care in econ/data:

- a bundle of n goods (quantities),
- a user's features or a product's attributes,
- a portfolio's weights across n assets,
- a document's word counts or an embedding.

Read Chapter 6 of Werner-Sotskov; Simon-Blume Chs. 10-11.

Exercises 6.2, 6.3, 6.4, 6.6, 6.7, 6.8

Vectors and the Euclidean space

Definition and notation

A **vector** $\mathbf{v}$ is an ordered n -tuple $(v_1, \dots, v_n)$ of real numbers called **coordinates**.

Notation: $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$.

The zero vector is $\mathbf{0} = (0, \dots, 0)$.

The i -th unit vector is $\mathbf{e}_i$ (a 1 in position i , zeros elsewhere).

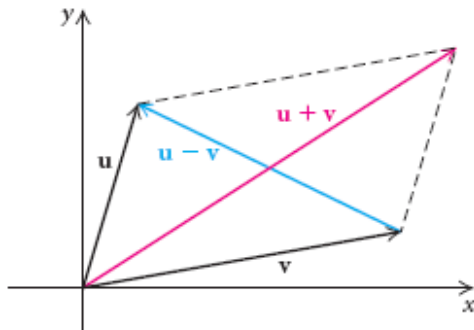
Operations:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \dots, u_n + v_n), \quad \lambda \mathbf{v} = (\lambda v_1, \dots, \lambda v_n).$$

These satisfy the usual commutative, associative, and distributive laws.

Sum and difference of two vectors

Note $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v}$.



Inner product and norm

The **inner product** (dot product) of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^n u_i v_i.$$

The **Euclidean norm** is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Properties:

1. $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. $\langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$
3. **Cauchy-Schwarz:** $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$
4. **Triangle inequality:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$

Example (platform revenue): hours watched per genre $\mathbf{u} = (500, 200, 50)$ and euro-per-hour rates $\mathbf{v} = (2, 3, 5)$ yield total revenue $\langle \mathbf{u}, \mathbf{v} \rangle = 2150$.

The Law of Cosines and angles

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\| \|\mathbf{v}\| \cos(\angle(\mathbf{u}, \mathbf{v})).$$

Equivalently,

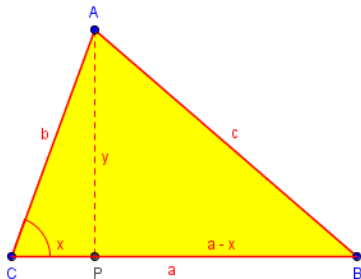
$$\cos(\angle(\mathbf{u}, \mathbf{v})) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}.$$

We prove that $c^2 = a^2 + b^2 - 2ab \cos(C)$:

By Pythagoras, $x^2 + y^2 = b^2$ and $(a-x)^2 + y^2 = c^2$.

Subtract to eliminate y^2 to get $c^2 = a^2 + b^2 - 2ax$.

Use $\cos C = x/b$.



Orthogonality and linear independence

Orthogonality

Definition: $\mathbf{u} \perp \mathbf{v}$ if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.

Angle example: For $\mathbf{u} = (2, -1, 3)$ and $\mathbf{v} = (5, -4, -1)$,

$$\langle \mathbf{u}, \mathbf{v} \rangle = 11, \quad \cos \angle(\mathbf{u}, \mathbf{v}) = \frac{11}{\sqrt{14}\sqrt{42}} \approx 0.4537,$$

so the angle is about 63° .

Geometric test: $\mathbf{u} \perp \mathbf{v}$ iff $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$.

Cauchy-Schwarz via completing the square

Assume nonzero $\mathbf{u}, \mathbf{v}$. For any real t ,

$$0 \leq \|\mathbf{u} - t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2.$$

Complete the square in t :

$$\|\mathbf{u}\|^2 - 2t \langle \mathbf{u}, \mathbf{v} \rangle + t^2 \|\mathbf{v}\|^2 = \|\mathbf{v}\|^2 \left(t - \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2} \right)^2 + \left(\|\mathbf{u}\|^2 - \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\|\mathbf{v}\|^2} \right).$$

Since the left side is ≥ 0 for all t , the second term must be ≥ 0 :

$$\langle \mathbf{u}, \mathbf{v} \rangle^2 \leq \|\mathbf{u}\|^2 \|\mathbf{v}\|^2.$$

Equality condition: equality holds iff $\|\mathbf{u} - t\mathbf{v}\|^2 = 0$ for $t = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}$, i.e., $\mathbf{u} = t\mathbf{v}$ (collinear vectors).



Linear dependence, independence, and bases

Linear combination: $\mathbf{u} = \sum_{i=1}^m \lambda_i \mathbf{v}_i$.

Linear independence: $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent if

$$\sum_{i=1}^m \lambda_i \mathbf{v}_i = \mathbf{0} \quad \Rightarrow \quad \lambda_1 = \dots = \lambda_m = 0.$$

Basis: Any n linearly independent vectors in $\mathbb{R}^n$ form a basis. Then every $\mathbf{u} \in \mathbb{R}^n$ has a **unique** representation $\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$.

Standard basis: $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

Subspaces, span, and dimension

For $E = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$, the **span** is

$$\text{span}(E) = \left\{ \sum_{i=1}^k \lambda_i \mathbf{v}_i : \lambda_i \in \mathbb{R} \right\}.$$

A **subspace** $V \subset \mathbb{R}^n$ is any span. A **basis of V** is a linearly independent set that spans V . The **dimension** of V is the size of any basis.

Exercise: Basis of $V = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Orthogonal complement and projection

Orthogonal complement

For a subspace $V \subset \mathbb{R}^n$,

$$V^\perp = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle = 0 \ \forall \mathbf{v} \in V\}.$$

Projection theorem

For any $\mathbf{y} \in \mathbb{R}^n$ and subspace V , there is a unique $\hat{\mathbf{y}} \in V$ with $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. We call $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto V .

Least squares view: With design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and response $\mathbf{y}$, the LS fit $\hat{\mathbf{y}}$ is the projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$.

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$



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Chapter 6: Matrices and determinants

A **matrix** $A \in \mathbb{R}^{m \times n}$ is a table with m rows, n columns. The (i, j) entry is a_{ij} .

We recall basic facts about matrices.

Why care: matrices express linear maps, data tables, network flows, input-output models, regressions, transformations, and more.

Read Werner-Sotskov Ch. 7; Simon-Blume Chs. 8-9.

Exercises 7.6, 7.9(b,c,d), 7.12, 7.14(a), 7.16, 7.18

Matrices

A table of numbers with m rows and n columns: $A \in \mathbb{R}^{m \times n}$. The (i, j) -th entry is denoted by a_{ij} .

Special matrices:

- zero matrix: $\mathbf{0}_{m \times n} \in \mathbb{R}^{m \times n}$
- identity matrix: $\mathbb{I}_n \in \mathbb{R}^{n \times n}$.
- **transposition**: $A^T \in \mathbb{R}^{n \times m}$, $(A^T)_{ij} = a_{ji}$.
- square matrix: $m = n$.
- symmetric matrix: square matrix such that $A = A^T$.
- diagonal matrix: $a_{ij} = 0$ if $i \neq j$.
- lower (upper) triangular: $a_{ij} = 0$ if $i < j$ ($i > j$).
- $\mathbf{v} \in \mathbb{R}^n$ is treated as $n \times 1$ matrix, $\mathbb{R}^n \simeq \mathbb{R}^{n \times 1}$.

Basic matrix operations

Addition and scalar multiplication are entrywise:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (\lambda A)_{ij} = \lambda a_{ij}.$$

They satisfy the usual commutative, associative, and distributive laws.

Matrix product: If $A \in \mathbb{R}^{m \times p}$ and $B \in \mathbb{R}^{p \times n}$, then $C = AB \in \mathbb{R}^{m \times n}$ with

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

Example:

$$\begin{pmatrix} 2 & 3 & 4 & 1 \\ 7 & -1 & 0 & 4 \end{pmatrix} \begin{pmatrix} 2 & 7 \\ 3 & -1 \\ 4 & 0 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 30 & 15 \\ 15 & 66 \end{pmatrix}.$$

Algebraic properties

Facts:

1. $(AB)C = A(BC)$
2. $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$
3. Generally **not** commutative: $AB \neq BA$
4. $AI_n = I_m A = A$ (dimensions must match)
5. $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(\lambda A)^T = \lambda A^T$, $(AB)^T = B^T A^T$

Remark: AA^T is always symmetric.

Matrix times vector

We treat $\mathbb{R}^n$ as column vectors $\mathbb{R}^{n \times 1}$. If $A \in \mathbb{R}^{m \times n}$ and $\mathbf{x} \in \mathbb{R}^n$ then $A\mathbf{x} \in \mathbb{R}^m$ is a linear combination of the columns of A with coefficients x_i :

$$A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n.$$

e.g. **Another look at the LS method:** $\mathbf{X} \in \mathbb{R}^{n \times d}$, $\mathbf{y} \in \mathbb{R}^n$

$$\text{minimize}_{\beta \in \mathbb{R}^d} \quad \|\mathbf{y} - \mathbf{X}\beta\|^2$$

Finds the closest point to $\mathbf{y}$ in the space spanned by the columns of $\mathbf{X}$.

Orthogonal projection

Theorem: Given a vector $\mathbf{y} \in \mathbb{R}^n$ and a subspace $V \subset \mathbb{R}^n$ there exists a unique $\hat{\mathbf{y}} \in V$ such that $\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$. Let $\mathbf{x}_1, \dots, \mathbf{x}_d$ be a basis of $V \subset \mathbb{R}^n$.

$\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_d$.

$\hat{\mathbf{y}} \in V$ means $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\lambda}$ for some $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)$.

$\mathbf{y} - \hat{\mathbf{y}} \in V^\perp$ means $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{y}}) = \mathbf{0}$.

The unique solution: $\boldsymbol{\lambda} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ giving

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Matrix inverse

A square matrix A is **invertible** if there exists a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

We want to show:

$$A \in \mathbb{R}^{n \times n} \text{ is invertible} \iff A\mathbf{x} = \mathbf{0}_n \text{ only for } \mathbf{x} = \mathbf{0}_n.$$

Two observations:

1. $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$ iff columns of A are lin. independent.
2. n independent vectors in $\mathbb{R}^n$ form a basis and so $\forall i = 1, \dots, n \exists \mathbf{b}_i$ such that $A\mathbf{b}_i = \mathbf{e}_i$.
3. This gives $B \in \mathbb{R}^{n \times n}$ such that $AB = \mathbb{I}_n$.

This is enough to show that the matrix $\mathbf{X}^\top \mathbf{X}$ on slide 46 is invertible.

Note: If $(\mathbf{X}^\top \mathbf{X})\mathbf{x} = \mathbf{0}$ then $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})\mathbf{x} = 0$ but this only possible if $\mathbf{x} = \mathbf{0}$.

Important spaces and rank

For $A \in \mathbb{R}^{m \times n}$:

- **Column space** (image) $\text{Im}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\} \subset \mathbb{R}^m$
- **Row space** $\text{Im}(A^T) \subset \mathbb{R}^n$
- **Kernel** (null space) $\ker(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\} \subset \mathbb{R}^n$
- **Rank** $\text{rank}(A) = \dim \text{Im}(A) = \dim \text{Im}(A^T)$

Orthogonality: Row space is orthogonal to $\ker(A)$.

Rank-nullity: $\text{rank}(A) + \dim \ker(A) = n$.

Try: $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

1	14	14	4
11	7	6	9
8	10	10	5
13	2	3	15



Orthogonal matrices

An $n \times n$ matrix A is called **orthogonal** if $AA^\top = I_n$.

Remark: If A is orthogonal, its row vectors (and also its column vectors) are pairwise orthogonal unit vectors.

Proof: Let $\mathbf{r}_i$ and $\mathbf{r}_j$ be the i -th and j -th rows of A . The (i, j) entry of $AA^\top$ is the scalar product $\mathbf{r}_i^\top \mathbf{r}_j$. If $AA^\top = I$, then $\mathbf{r}_i^\top \mathbf{r}_j = 0$ for $i \neq j$ (orthogonality) and $\mathbf{r}_i^\top \mathbf{r}_i = \|\mathbf{r}_i\|^2 = 1$ (unit length). The same holds for columns using $A^\top A = I_n$.

Example:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Singular value decomposition (SVD)

Singular value decomposition

For every $A \in \mathbb{R}^{m \times n}$ there exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ and a “diagonal” matrix $\Lambda \in \mathbb{R}^{m \times n}$ such that

$$A = U\Lambda V^T.$$

If $m \leq n$ then Λ has m “diagonal” entries.

We can assume $\Lambda_{11} \geq \dots \geq \Lambda_{mm} \geq 0$.

These diagonal entries are called **singular values** of A .

Spectral decomposition

Singular value decomposition

Suppose $A \in \mathbb{R}^{n \times n}$ is symmetric. Then there exist orthogonal $U \in \mathbb{R}^{n \times n}$ and a diagonal Λ such that

$$A = U\Lambda U^{\top}.$$

Denote $(\lambda_1, \dots, \lambda_n) = \text{diag}(\Lambda)$.

We take $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$.

Elementary matrix operations

Column or row operations:

1. **Swap** two columns (or rows)
2. **Scale** a column (or row) by $\lambda \neq 0$
3. **Add** a multiple of one column (or row) to another

Each is implemented by multiplying by a suitable elementary matrix on the right (for column ops) or left (for row ops). Useful for Gaussian elimination and determinant computation.

To see what is happening, let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and multiply from the right by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}$, or $\begin{bmatrix} 1 & 0 \\ \lambda & 1 \end{bmatrix}$.

Determinants: definition

For $A \in \mathbb{R}^{n \times n}$, let A_{ij} be the submatrix with row i and column j removed. Define recursively

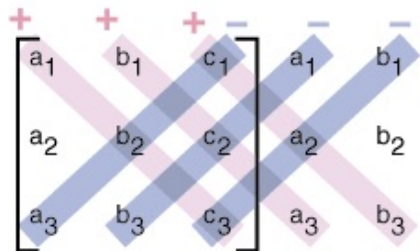
$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}), \quad \det([a_{11}]) = a_{11}.$$

For $n = 2$: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$

For $n = 3$:

$$\det(A) = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}.$$

A memory aid for $n = 3$



$$\det A = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1)$$

Cofactor expansion and properties

Cofactor expansion: expand by any row i or any column j :

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}).$$

Properties:

1. $\det(A) = \det(A^T)$
2. If A is triangular, $\det(A)$ is the product of diagonal entries
3. $\det(AB) = \det(A) \det(B)$
4. Swapping two rows (or columns) flips the sign of $\det$
5. Scaling a row (or column) by λ scales $\det$ by λ
6. Adding a multiple of one row to another leaves $\det$ unchanged
7. $\det(A) = 0$ iff rows (or columns) are linearly dependent

Determinants by elimination

Use row operations (keeping track of determinant changes) to reach triangular form.

$$\begin{vmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 1 & 2 & 3 \end{vmatrix} \rightarrow \begin{vmatrix} 1 & 2 & 0 \\ 0 & -6 & 1 \\ 0 & 0 & 3 \end{vmatrix} = (-6) \cdot 3 = -18.$$

First: $r_2 \leftarrow r_2 - 3r_1$. Then: $r_3 \leftarrow r_3 - r_1$.

Geometric meaning: $|\det(A)|$ is the area/volume scaling of the linear map $x \mapsto Ax$ (and its sign encodes orientation).

Linear systems and Cramer's rule

A system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{n \times n}$, unknown $\mathbf{x} \in \mathbb{R}^n$, and data $\mathbf{b}$. If A is **nonsingular** ($\det A \neq 0$), the solution is unique.

Cramer's rule: Let $A_j(\mathbf{b})$ be A with column j replaced by $\mathbf{b}$. Then

$$x_j = \frac{\det A_j(\mathbf{b})}{\det A}, \quad j = 1, \dots, n.$$

Note: Great for theory, not used for large-scale computation.

Example:

$$A = \begin{pmatrix} 2 & 3 & 5 \\ 1 & 0 & 2 \\ -1 & -4 & 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = \left(\frac{75}{8}, -\frac{63}{16}, -\frac{19}{16} \right).$$

Linear mappings

A mapping $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **linear** if

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A(\mathbf{x}_1) + A(\mathbf{x}_2), \quad A(\lambda \mathbf{x}) = \lambda A(\mathbf{x}).$$

Then there exists an $m \times n$ matrix (also denoted A) with $A(\mathbf{x}) = A\mathbf{x}$.

Columns as images: $A(\mathbf{e}_i) = \mathbf{a}_i$ (the i -th column), and $A(\mathbf{x}) = \sum_i x_i A(\mathbf{e}_i)$.

Examples: scalings, rotations, reflections, projections, feature maps in ML, Leontief input-output in economics.

Two simple linear maps

1. Reflection across the y -axis:

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}.$$

2. Rotation by 45° counterclockwise:

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

Check the images of $\mathbf{e}_1$ and $\mathbf{e}_2$ to see the action.

Inverse matrix and properties

A square A is invertible if A^{-1} exists with $AA^{-1} = A^{-1}A = I_n$.

Properties:

1. $(A^{-1})^{-1} = A$
2. $(AB)^{-1} = B^{-1}A^{-1}$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(\lambda A)^{-1} = \lambda^{-1}A^{-1}$ for $\lambda \neq 0$
5. $\det(A^{-1}) = 1/\det(A)$

Solve $A\mathbf{x} = \mathbf{b}$ by $\mathbf{x} = A^{-1}\mathbf{b}$ when A is invertible.

Computing inverses

Cofactor formula: If A is nonsingular,

$$A^{-1} = \frac{1}{\det A} C(A)^T, \quad C(A)_{ij} = (-1)^{i+j} \det(A_{ij}).$$

For 2×2 :

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

For larger n , numerical methods use elimination (LU), not cofactors.

Example: Input-output model

Let a_{ij} be units of good i needed to produce 1 unit of good j . Put $A = (a_{ij})$. Let $\mathbf{x}$ be total output and $\mathbf{y}$ the vector of final demand.

Accounting identity: output = internal demand + final demand

$$\mathbf{x} = A\mathbf{x} + \mathbf{y} \quad \Leftrightarrow \quad (I_n - A)\mathbf{x} = \mathbf{y} \quad \Rightarrow \quad \mathbf{x} = (I_n - A)^{-1}\mathbf{y}$$

provided $I_n - A$ is invertible.

Interpretation: $(I - A)^{-1} = I + A + A^2 + \dots$ accumulates direct, indirect, and higher-order input needs when it converges.

A triangular example

Theorem: If A is strictly upper triangular (zeros on and below diagonal), then $A^n = 0$ and

$$(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{n-1}.$$

Check $(I - A)(I + A + \cdots + A^{n-1}) = I - A^n = I$.

Example:

$$A = \begin{pmatrix} 0 & 3 & 5 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 1 \\ 7 \\ 4 \end{pmatrix} \Rightarrow \mathbf{x} = (66, 15, 4).$$